Fourteen-position prime-norm schedule and fixed-radix measurement

This report records two distinct findings for the secp256k1 fixed-base τ-radix evaluator: an exact integer search for a lower direct point payload, and a Tier 1 native throughput comparison for the fixed `371−τ` schedule. The latter does not measure the ordered schedule selected by the screen.

Search and correctness evidence

The screen enumerates all 50 prime norms `T²−3T+3` for `T=300…600`, checks residue/sign/unit covariance and exact radius contractions, then runs four declared coordinate-descent starts (`371`, `380`, `394`, `400`) with choices from `T=330…430`. The selected local schedule is `394−τ, 380−τ, 380−τ, (371−τ)×11`. It lowers the computed direct point payload from 22,313,920 to 22,267,136 bytes, a 46,784-byte reduction. The coordinate search is local; no global optimum over ordered schedules is claimed. Re-running `screen14.py` reproduced `screen14.json` byte-for-byte.

`replay14.py` reverses the quotient steps and checks the terminal table and point action against independent binary secp256k1 multiplication on 164 scalars. Its regenerated output matched `replay14.json` byte-for-byte. The conditional terminal-radius theorem and its axioms audit are recorded in `OrderedPrimeNorm14Bound.lean`, `lean-check.log`, and `lean-fresh-check.log`; the selected-digit proof and concrete group-action bridge remain explicitly open.

Native measurement

The frozen panel compares fixed `371−τ` at 14 positions with the fixed 15-position `268−τ` reference on 12,288 same-point outputs, with 16 alternating repetitions, 48 paired comparisons, and adjacent A/A controls. The committed independent audit replayed every arm and gate; regenerated `analyze_fixed14_tier1.py` output matched the committed analysis byte-for-byte.

Measure	Fixed 14 (`371−τ`)	Reference 15 (`268−τ`)
Mean online time per 12,288 outputs	135.822 ms	138.354 ms
Mean table preparation	3,953.993 ms	2,103.098 ms
Retained native table	26,035,988 bytes	12,336,756 bytes

Reference/candidate online time was 1.020033×, with the committed repeat- bootstrap 95% interval [1.016715, 1.023710]. The interval is an instance-scoped Tier 1 result; the RunPod CPU affinity does not establish physical-host isolation. Preparation and retained-memory costs are higher for the measured fixed-14 evaluator. The ordered schedule's native cost and throughput remain unmeasured.

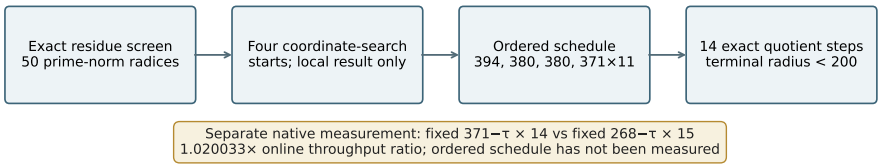


Figure 1: Search and measurement scopes

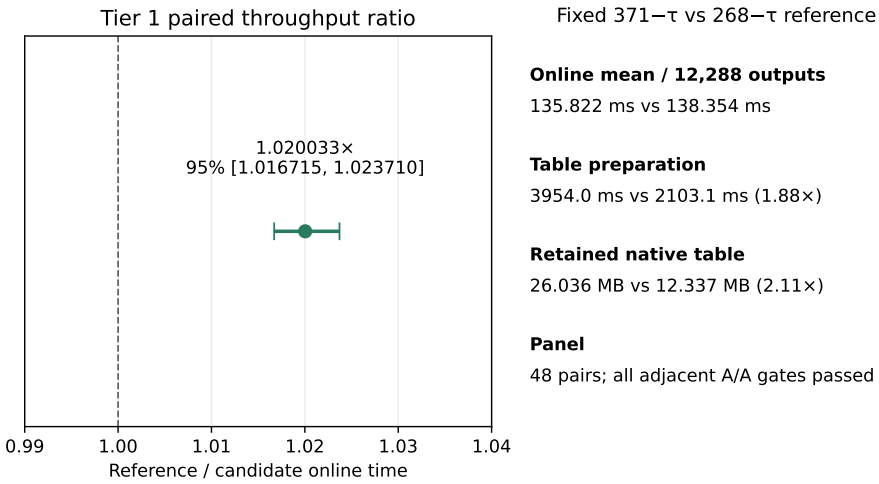


Figure 2: Tier 1 paired throughput result

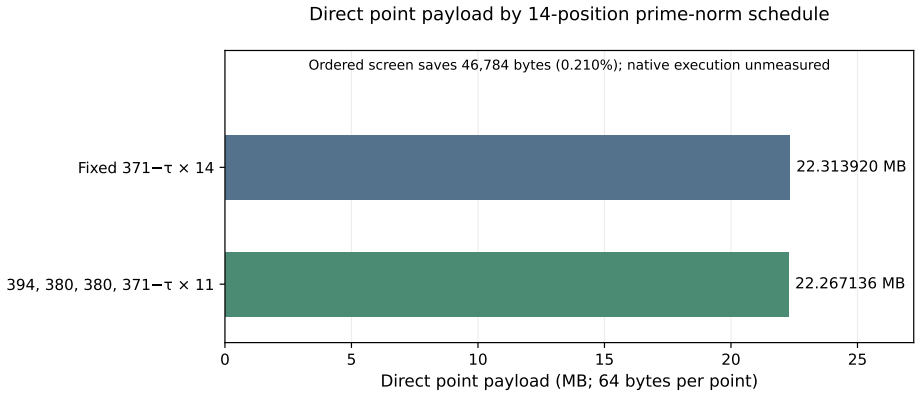


Figure 3: Direct point-payload screen

Visuals and graph scope

`visuals.py` is the editable source for the search-and-measurement diagram and both quantitative figures: `search-and-measurement.svg`, `tier1-throughput.svg`, and `ordered-payload.svg`. The source reads the committed screen and analysis JSON; regenerating it produces SVG and PDF figures. The existing curve/isogeny route graphs were checked: this result introduces no curve, map, subgroup, or route edge. The measured comparison and payload screen are recorded in the experiment-specific graph set here.

Build the report PDF with:

```
MPLCONFIGDIR=/tmp/mpl-cache XDG_CACHE_HOME=/tmp/font-cache python3 visuals.py
SOURCE_DATE_EPOCH=0 pandoc REPORT-2026-10-19.md --resource-path=.:figures --pdf-engi
  -V mainfont="DejaVu Serif" -V monofont="DejaVu Sans Mono" \
  -o REPORT-2026-10-19.pdf
```

The fixed-15 frontier README links this record as the measured and screened follow-on. Relevant evidence remains in `README.md`, `screen14.json`, `replay14.json`, `tier1-fixed14-analysis.json`, the frozen raw panel directory, and `radix136531-panel.tar.gz`.